

ADVANCED WATER SAFETY DATA PACK

Advanced Data Analysis Activities | Year 9-10 HPE

Four activities using real Victorian and Australian data.
Trend analysis, CALD argument, global comparison,
and diving & spinal injury data. All answers evidence-based.

Activity A — Victorian Drowning Rate Trend Analysis

Data source: Life Saving Victoria Victorian Drowning Report 2023-24

The table below shows Victorian drowning death totals 2019-20 to 2023-24. Population approx. 6.9 million throughout. Use this data to complete the activities below.

Year	Deaths	Rate per 100,000	% change from prior year
2019-20	44		
2020-21	37		
2021-22	42		
2022-23	47		
2023-24	54		

Q.A1 [3 marks]

Calculate the Victorian drowning rate per 100,000 population for each year. Show your working in the table above. Then describe the overall trend in two sentences, including the percentage increase from 2020-21 (lowest) to 2023-24 (highest).

Q.A2 [3 marks]

The national Australian drowning rate in 2023-24 was approximately 1.2 per 100,000. Victoria has approximately 6.9 million people. Calculate: (a) the expected number of Victorian deaths at the national rate, and (b) what this comparison tells us about Victoria's relative performance.

Activity B — CALD Over-Representation Argument Analysis

Data: CALD communities account for approximately 40% of Victorian drowning deaths but approximately 28% of the population (LSV 2023-24)

Q.B1 [4 marks]

Calculate the CALD over-representation ratio. Then explain TWO structural reasons — not individual ones — that explain this disproportionality. For each reason, describe a specific prevention response that addresses the structural cause.

Q.B2 [3 marks]

A colleague argues: "CALD community members just need better water safety education." Explain why this argument is insufficient, using the concept of structural determinants of health. What would a more complete response look like?

Activity C — Global Comparison

Research task: Use royallifesaving.com.au, who.int, or redcross.org for this activity.

For this activity, research ONE other country's drowning statistics and prevention approach. Compare it to Victoria using evidence.

Q.C1 [3 marks]

Identify the annual drowning rate per 100,000 population for ONE country other than Australia. State your source with URL. Calculate whether this country's rate is higher or lower than Victoria's 2023-24 rate and by what factor.

Hint: Try: WHO Global Drowning Report (who.int) or RLSS UK (rlss.org.uk)

Q.C2 [4 marks]

Identify ONE prevention strategy used in your chosen country that differs meaningfully from Australia's approach. Evaluate: (a) the evidence base for this strategy, (b) whether the contextual differences between that country and Victoria would affect its success here, and (c) whether you would recommend adapting it for Victoria.

Activity D — Diving & Spinal Injury Data

Data sources: AIHW Diving Injury Report | Royal Perth Hospital spinal injury study | RLSSA

Unlike drowning data (reported annually), spinal injury data is compiled across multi-year studies. Use the data below alongside Victorian drowning data from Activities A-C.

Statistic	Value	Source
% of all spinal cord injuries from unsafe diving	10%	AIHW
% of quadriplegia cases from diving	20%	AIHW
% of aquatic spinal injury patients who are male	95%	Royal Perth Hospital review
% of patients under 30 years	75%+	Royal Perth Hospital review
% of patients with permanent disability	66%	Royal Perth Hospital review
Highest incidence age group	10-14 years	AIHW
Post-COVID drowning change: 15-20 year olds	+34%	RLSSA

Prevention context:

"Feet First, First Time" — always enter unknown water feet-first; check depth on every visit regardless of prior experience. Inland waterways (rivers, dams, lakes) carry the highest drowning risk of any environment in Australia and present additional hazards: cold water shock, poor visibility, and remote rescue delay.

Q.D1 [3 marks]

Compare the demographic distribution of Victorian drowning victims with the demographic distribution of aquatic spinal injury patients. What does this comparison reveal about the limitations of using drowning data alone to understand dangerous water behaviour among young Australians?

Q.D2 [3 marks]

If approximately 100 new aquatic spinal cord injury cases occur in Australia per year and 95% affect males aged 10-30 (approx. 3.2 million in that population): (a) Calculate the annual rate per 100,000 males aged 10-30. (b) Compare this to the Victorian drowning rate of 0.78 per 100,000. (c) Identify two significant limitations of this calculation.

Q.D3 [4 marks]

The post-COVID data shows a 34% increase in drowning among 15-20 year olds. Propose THREE possible explanations. For each, identify: (a) what evidence would support it, and (b) what prevention response would follow if it were the primary cause.
